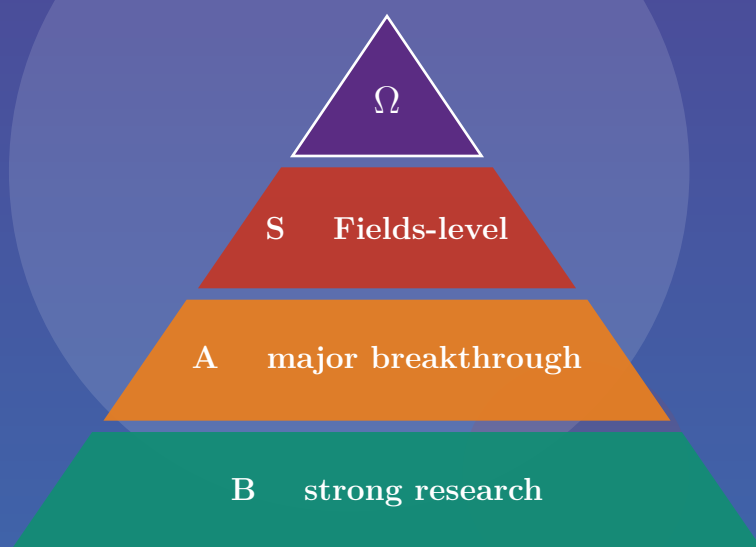


The Atlas of Hard Problems

100 open problems in mathematics, ranked from *beyond Fields* down to *strong research*, with an honest map of which ones raw reasoning, discrete mathematics and a proof assistant could plausibly reach.



Special Project · October 2026

Written by Claude for the project's mathematician. All rankings and odds are my own judgement.

The short version

Five things to know before reading the list

1. **There is no official ranking.** The tiers and scores below are my judgement, informed by how the mathematical community treats each problem: prizes attached, what partial results earned, and how many strong people have tried.
2. **The Fields Medal rewards a body of work by someone under 40, not one problem.** Here, *Fields-level* means a result that on its own would make an overwhelming case for the medal. Above that sits Tier Ω : problems whose solution would change what mathematics is about (seven are Millennium or Millennium-class).
3. **“Very hard” and “nobody has tackled it” rarely go together.** Famous problems are famous because many people failed at them. Hard problems nobody has tried exist mostly as *freshly posed* questions (like our own Lucas conjecture) and as lightly studied Erdős problems.
4. **Intelligence is not the only bottleneck.** For the top problems there are *barrier theorems*: proofs that whole families of methods cannot work (page 9). Any solution must find a door that is not yet on any map.
5. **There is a real sweet spot:** purely discrete problems whose recent history contains *short, idea-driven proofs* (sensitivity in 6 pages, cap sets in 4, union-closed sets in about 8). That is where our toolkit of reasoning, discrete mathematics and Lean is strongest. My picks are on page 10.

How each problem is scored

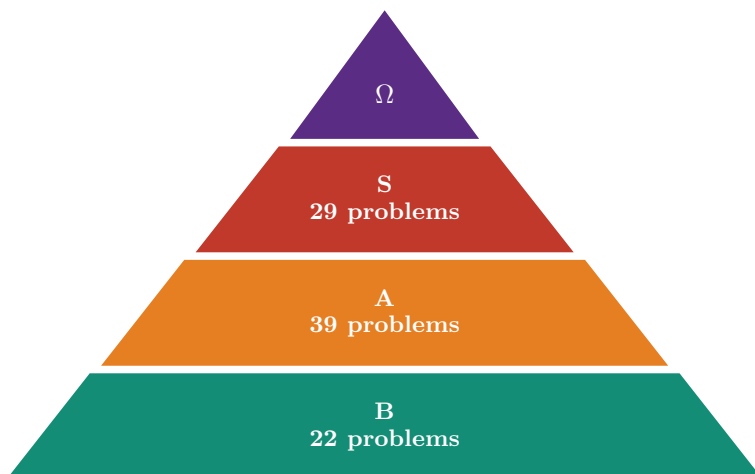
Tier (colour) records how big the *reward* would be:

- Ω : beyond Fields
- S: Fields-level
- A: major breakthrough
- B: strong research problem

Difficulty 6 7 8 9 10: how far today’s methods are from a solution (10 = no plausible route is known).

Raw-reasoning fit ■■■■ to ■■■■■: how likely progress is to come from a clean conceptual idea plus careful discrete mathematics, the kind of argument we could check in Lean, rather than from decades of heavy machinery or massive computer search.

Tier and difficulty are different axes. Proving infinitely many Fermat primes would be fiendishly hard (high D) but would change little; the cap-set bound was “only” Tier A in reward yet fell to a 4-page idea.



Ω (10). Millennium-class. Riemann, P vs NP, Navier–Stokes, BSD, Hodge, twin primes. Prize: \$1M, plus immortality.

S (29). Would dominate a Fields citation: abc, Collatz, Kakeya, Jacobian, Frege lower bounds. Avg. fit 2.1/5.

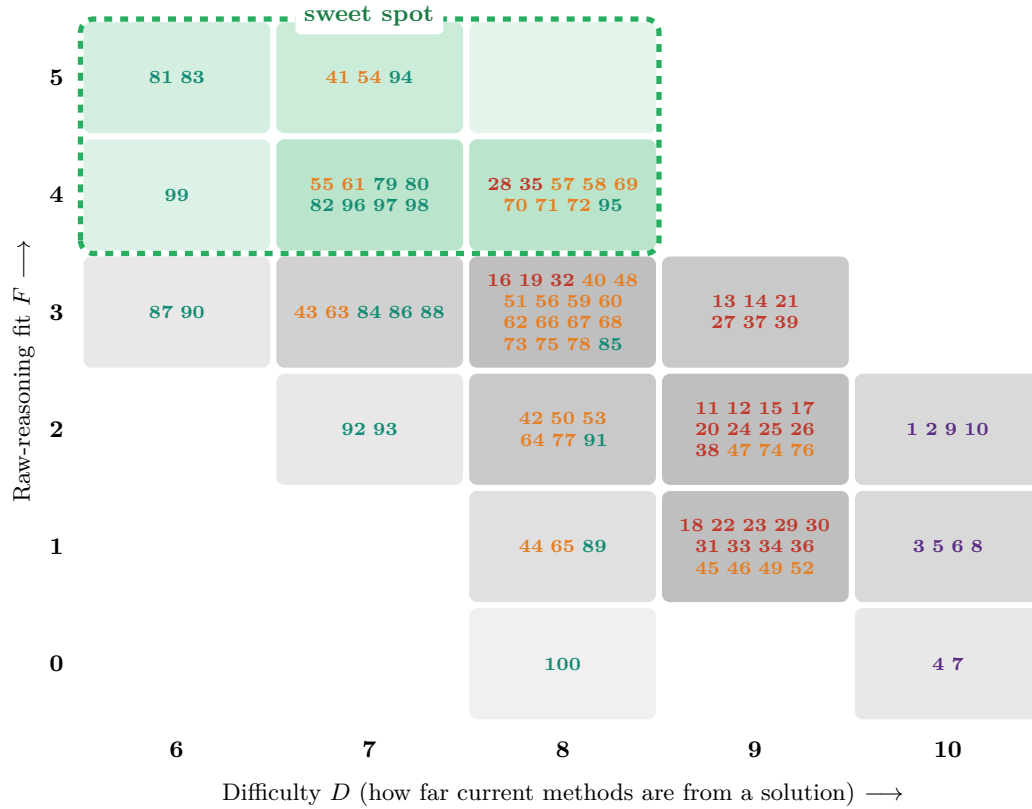
A (39). An Annals-level paper or a major prize: union-closed sets, $\zeta(5)$, Hadwiger–Nelson. Avg. fit 2.8/5.

B (22). Famous among specialists: Seymour’s second neighbourhood, Rota’s basis, Erdős–Gyárfás. Avg. fit 3.3/5.

Notice: average fit *rises* as you go down the pyramid (Ω : 1.2/5). The most famous problems are the least likely to fall to one clever idea, which is part of why they are still famous.

1 The map: difficulty against fit

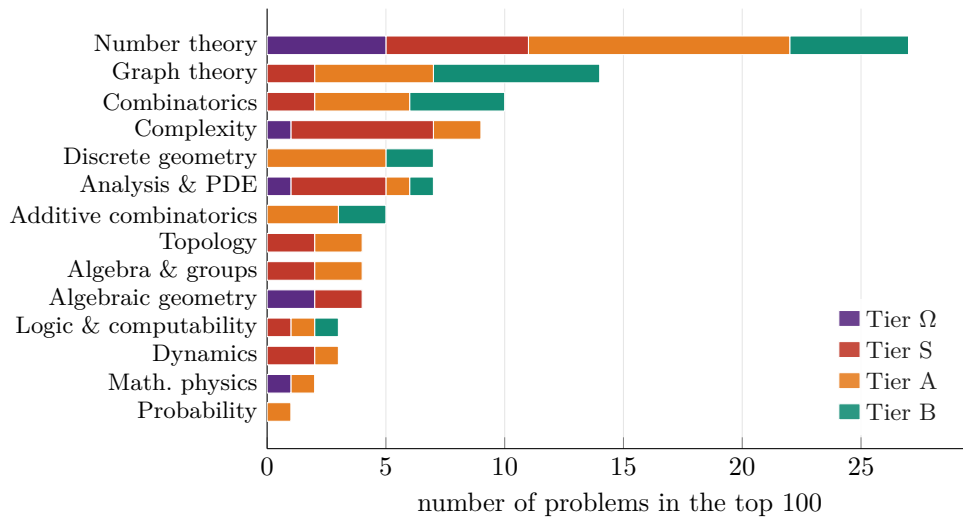
Each number is a problem from the list in Section 3, coloured by tier. Cells shaded green form the **sweet spot** (fit ≥ 4 , difficulty ≤ 8): hard enough to matter, but the kind of problem that recently fell to short ideas.



Reading the map

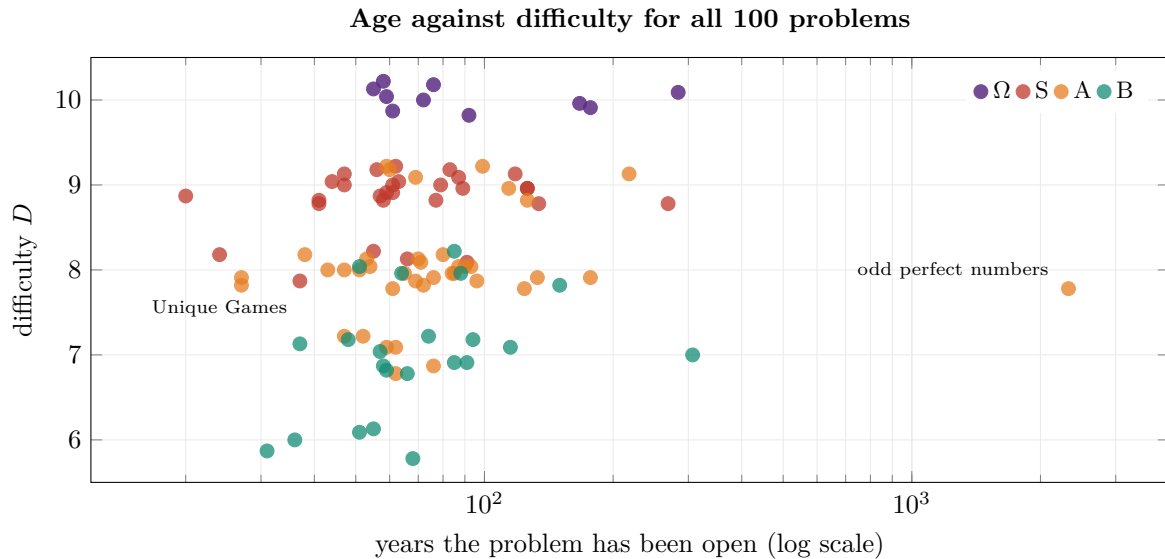
- **Top-left (green):** the realistic targets. Union-closed sets (41), Černý (54), Seymour (81), Erdős–Gyárfás (83), the $1/3$ – $2/3$ conjecture (94), Sunflower (35), Erdős–Hajnal (28).
- **Bottom-right:** problems that need *new theories* (Langlands, Yang–Mills, Hodge). No single idea, however clever, is known to suffice.
- **Bottom-middle:** problems that need *huge computation* ($R(5, 5)$, projective plane of order 12, $BB(6)$). They are off-limits by your rule of no heavy computation.

Where the hard problems live



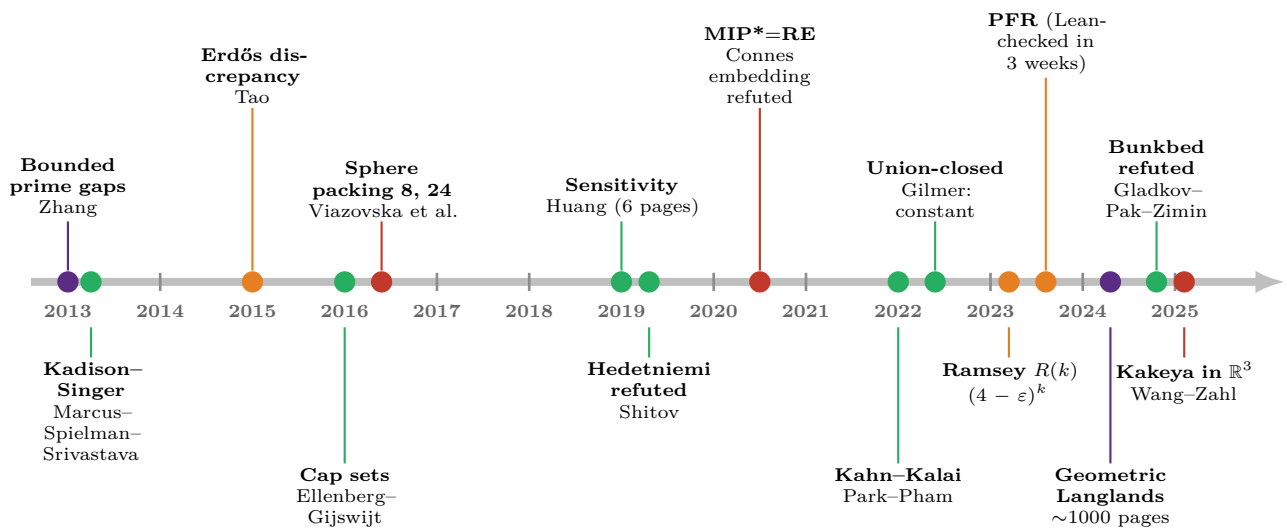
Number theory dominates the top tier: its statements are simple and its depth is extreme. Graph theory and combinatorics dominate the lower tiers, which are also where the fit is highest.

Old is not the same as hard



Problems from the 1960s–80s sit at every difficulty level. Age measures how long people have been stuck, not how deep the problem is. Collatz (1937) is old and famous, but its *depth* is unclear: Conway showed that a generalisation is undecidable, so it may be hard for reasons unrelated to deep structure.

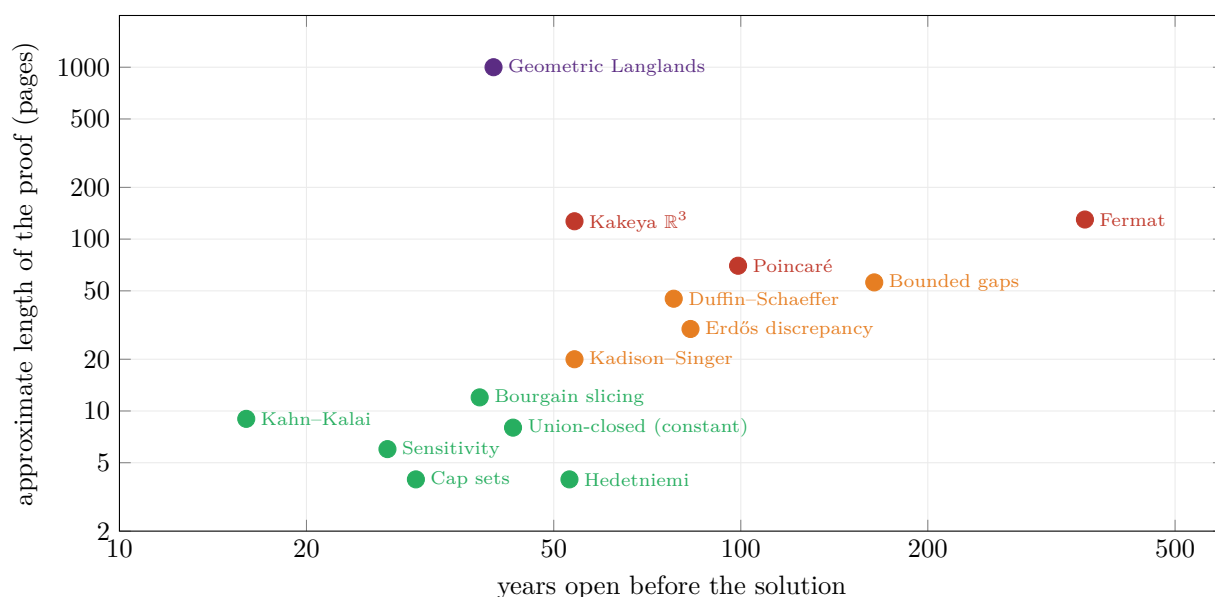
2 Lessons from the problems that fell (2013–2025)



- short idea-driven proof (or a short counterexample)
- a substantial new method
- a long, deep proof
- a programme spanning many people and years.

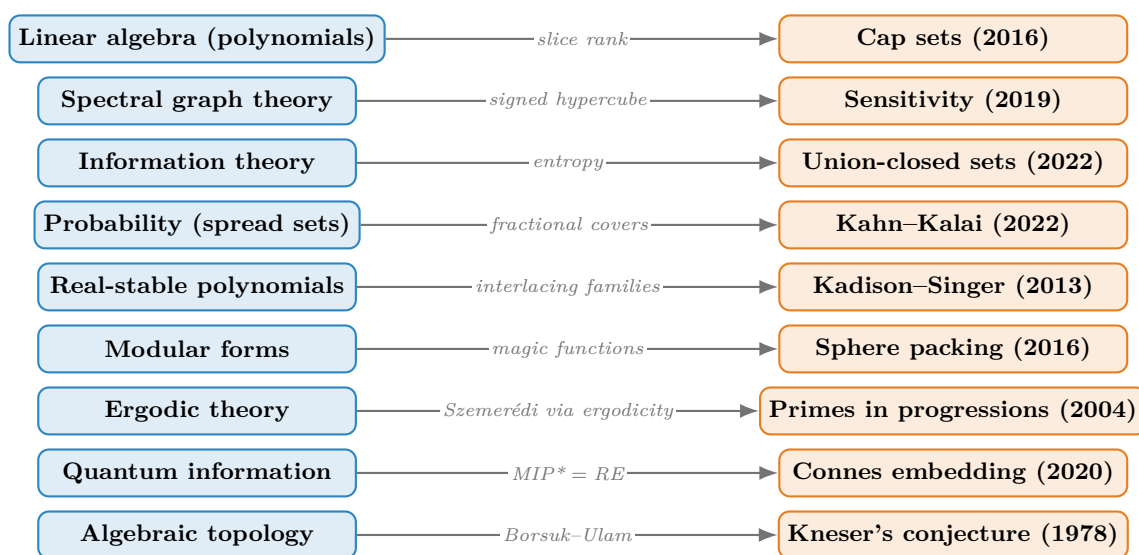
Proof length is almost uncorrelated with how long a problem was open

Many decades-old problems fell to proofs under 10 pages



Colours as in the timeline: **green** short idea, **orange** new method, **red** long deep proof, **purple** programme. Page counts are rough (main paper only; Perelman’s three preprints, Wiles plus Taylor–Wiles). The lesson is robust: **the green cluster at the bottom is real**. In combinatorics and discrete mathematics, problems open for 30–50 years have repeatedly been solved by one new viewpoint, written up in a few pages.

Where the winning ideas came from: transplants between fields



This is the strongest argument for an interdisciplinary attack. None of these proofs came from grinding harder inside the problem’s own field. My one real structural advantage is breadth: I can search systematically for a transplant, asking which tool from field X has the right shape for problem Y .

3 The 100 problems

#	Problem	Since	Best known / status (2026)	D	Fit
Tier Ω — beyond Fields (would reshape mathematics)					
1	Riemann Hypothesis Number theory	1859	$> 41\%$ of zeros proved on the line; 10^{13} zeros checked	10	■■■■■
2	P vs NP Complexity	1971	Three barriers: relativization, natural proofs, algebrization	10	■■■■■
3	Navier–Stokes regularity Analysis & PDE	1934	An averaged NS equation blows up (Tao 2016), so energy methods alone fail	10	■■■■■
4	Yang–Mills mass gap Math. physics	1954	No interacting 4D quantum field theory has been rigorously built	10	■■■■■
5	Hodge conjecture Algebraic geometry	1950	Known for $(1, 1)$ -classes (Lefschetz); open in general	10	■■■■■
6	Birch–Swinnerton-Dyer Number theory	1965	Analytic rank ≤ 1 (Gross–Zagier, Kolyvagin); positive proportion of curves	10	■■■■■
7	Langlands functoriality Number theory	1967	Geometric Langlands proved (2024, ~ 1000 pp); arithmetic case wide open	10	■■■■■
8	Grothendieck’s standard conjectures Algebraic geometry	1968	Special cases only; would give a theory of motives	10	■■■■■
9	Twin prime conjecture Number theory	1849	Gaps ≤ 246 infinitely often (Zhang, Maynard, Polymath); parity barrier	10	■■■■■
10	Goldbach conjecture (strong) Number theory	1742	Weak (ternary) Goldbach proved (Helfgott 2013); checked to $4 \cdot 10^{18}$	10	■■■■■
Tier S — Fields-level (a solution is a career-defining, prize-certain result)					
11	abc conjecture (accepted proof) Number theory	1985	IUT claim (Mochizuki 2012) disputed by Scholze–Stix (2018)	9	■■■■■
12	Collatz conjecture Number theory	1937	Almost all orbits attain almost bounded values (Tao 2019)	9	■■■■■
13	Frege lower bounds (Frege vs EF) Complexity	1979	No superpolynomial Frege lower bound; tied to our naming project	9	■■■■■
14	Explicit circuit lower bounds Complexity	1949	Best $\approx 3.1n$ for circuits, $\approx n^3$ for formulas	9	■■■■■
15	VP vs VNP Complexity	1979	Permanent needs determinants of size $\geq n^2/2$ (Mignon–Ressayre)	9	■■■■■
16	Unique Games Conjecture Complexity	2002	2-to-2 Games Theorem (Khot–Minzer–Safra 2018)	8	■■■■■
17	Matrix multiplication $\omega = 2$ Complexity	1969	$\omega < 2.3714$; barriers for laser-method variants	9	■■■■■
18	Smooth 4D Poincaré conjecture Topology	1982	Topological case: Freedman (1982); smooth case open	9	■■■■■
19	Kakeya conjecture, $n \geq 4$ Analysis & PDE	1971	Solved in $\mathbb{R}^3$ (Wang–Zahl 2025)	8	■■■■■
20	Fourier restriction conjecture Analysis & PDE	1967	Polynomial partitioning and decoupling give partial ranges	9	■■■■■
21	Jacobian conjecture Algebra & groups	1939	Reduced to cubic homogeneous maps; many false proofs	9	■■■■■
22	Inverse Galois problem Algebra & groups	1892	All solvable groups (Shafarevich); M_{23} the only open sporadic	9	■■■■■
23	Hilbert’s 12th problem Number theory	1900	Totally real fields: Dasgupta–Kakde (2021+)	9	■■■■■
24	Lindelöf hypothesis Number theory	1908	$\zeta(\frac{1}{2} + it) \ll t^{13/84+\varepsilon}$ (Bourgain 2017)	9	■■■■■
25	Elliott–Halberstam conjecture Number theory	1968	Level $1/2$ (Bombieri–Vinogradov); beyond for smooth moduli	9	■■■■■
26	Chowla and Sarnak conjectures Number theory	1965	Log-averaged two-point Chowla (Tao 2016)	9	■■■■■
27	Hadwiger’s conjecture Graph theory	1943	True for $t \leq 6$; $O(t \log \log t)$ colours (Delcourt–Postle)	9	■■■■■
28	Erdős–Hajnal conjecture Graph theory	1989	P_5 -free case proved (Nguyen–Scott–Seymour 2023)	8	■■■■■
29	Tate conjecture Algebraic geometry	1963	Divisors on abelian varieties (Faltings); K3 surfaces	9	■■■■■

#	Problem	Since	Best known / status (2026)	D	Fit
30	Resolution of singularities, char p Algebraic geometry	1964	Known up to dimension 3	9	■ ■ ■ ■ ■
31	Novikov / Baum–Connes Topology	1965	Hyperbolic and amenable groups; groupoid counterexamples	9	■ ■ ■ ■ ■
32	Invariant subspace problem Analysis & PDE	1935	False on some Banach spaces (Enflo, Read); Hilbert space open	8	■ ■ ■ ■ ■
33	3D Euler: smooth blow-up? Analysis & PDE	1757	Blow-up with a boundary (Chen–Hou 2022, computer-assisted)	9	■ ■ ■ ■ ■
34	Hilbert’s 16th (limit cycles) Dynamics	1900	No uniform bound even for quadratic vector fields	9	■ ■ ■ ■ ■
35	Sunflower conjecture Combinatorics	1960	$(C \log k)^k$ -type bounds via spread sets (ALWZ 2019, Rao)	8	■ ■ ■ ■ ■
36	MLC: Mandelbrot set locally connected Dynamics	1985	Known at many parameters (Yoccoz); open in general	9	■ ■ ■ ■ ■
37	Diagonal Ramsey growth rate Combinatorics	1947	$R(k) \leq (4 - \varepsilon)^k$ (Campos–Griffiths–Morris–Sahasrabudhe 2023)	9	■ ■ ■ ■ ■
38	Hilbert’s 10th problem over $\mathbb{Q}$ Logic & computability	1970	Undecidable over all rings of integers (Koymans–Pagano 2024)	9	■ ■ ■ ■ ■
39	Quantum PCP conjecture Complexity	2006	NLTS theorem (Anshu–Breuckmann–Nirkhe 2022)	9	■ ■ ■ ■ ■
Tier A — major breakthrough (top-journal paper, major prize)					
40	Erdős arithmetic-progression conj. Additive combinatorics	1936	$k = 3$ proved (Bloom–Sisask 2020); Erdős offered \$5000	8	■ ■ ■ ■ ■
41	Union-closed sets (Frankl) Combinatorics	1979	Constant ≈ 0.38 (Gilmer 2022 and successors); $1/2$ open	7	■ ■ ■ ■ ■
42	Chromatic number of the plane Discrete geometry	1950	$5 \leq \chi \leq 7$ (de Grey 2018, computer-assisted)	8	■ ■ ■ ■ ■
43	Lonely runner conjecture Number theory	1967	Proved for up to 7 runners; recent computer-assisted extensions	7	■ ■ ■ ■ ■
44	Odd perfect numbers Number theory	c. 300 BC	Any example exceeds 10^{1500} (Ochem–Rao)	8	■ ■ ■ ■ ■
45	Primes of the form $n^2 + 1$ Number theory	1912	$n^2 + 1$ has ≤ 2 prime factors infinitely often (Iwaniec)	9	■ ■ ■ ■ ■
46	Legendre: prime in $(n^2, (n+1)^2)$ Number theory	1808	Not implied even by RH; gaps $x^{0.525}$ (Baker–Harman–Pintz)	9	■ ■ ■ ■ ■
47	Irrationality of $e + \pi$, $e\pi$, γ Number theory	1900	At least one of $e + \pi$, $e\pi$ is transcendental	9	■ ■ ■ ■ ■
48	Irrationality of $\zeta(5)$ Number theory	1979	One of $\zeta(5), \zeta(7), \zeta(9), \zeta(11)$ irrational (Zudilin 2001)	8	■ ■ ■ ■ ■
49	Schanuel’s conjecture Number theory	1966	Would give algebraic independence of e and π	9	■ ■ ■ ■ ■
50	Littlewood conjecture Number theory	1930	Exceptions have dimension 0 (Einsiedler–Katok–Lindenstrauss)	8	■ ■ ■ ■ ■
51	Lehmer’s Mahler-measure problem Number theory	1933	Lehmer’s 1.17628... still the smallest known value	8	■ ■ ■ ■ ■
52	Artin’s primitive-root conjecture Number theory	1927	True under GRH (Hooley 1967)	9	■ ■ ■ ■ ■
53	Gauss circle / Dirichlet divisor Number theory	1849	Exponent ≈ 0.3145 vs. conjectured $1/4$	8	■ ■ ■ ■ ■
54	Černý conjecture Combinatorics	1964	Best bound $\approx n^3/6$; conjecture $(n-1)^2$	7	■ ■ ■ ■ ■
55	Graceful tree conjecture Graph theory	1964	Ringel’s conjecture proved (Montgomery–Pokrovskiy–Sudakov 2020)	7	■ ■ ■ ■ ■
56	Graph reconstruction conjecture Graph theory	1942	Checked to 13 vertices; true for almost all graphs	8	■ ■ ■ ■ ■
57	Cycle double cover conjecture Graph theory	1973	Reduced to snarks	8	■ ■ ■ ■ ■
58	Tutte’s 5-flow conjecture Graph theory	1954	Every bridgeless graph has a 6-flow (Seymour 1981)	8	■ ■ ■ ■ ■
59	List colouring conjecture Graph theory	1975	Bipartite case (Galvin 1995); asymptotically (Kahn 1996)	8	■ ■ ■ ■ ■

#	Problem	Since	Best known / status (2026)	D	Fit
60	Erdős unit-distance problem Discrete geometry	1946	Upper bound $O(n^{4/3})$ unchanged since 1984	8	■■■■■
61	Heilbronn triangle problem Discrete geometry	1950	$n^{-8/7-1/2000}$ (Cohen–Pohoata–Zakharov 2023)	7	■■■■■
62	Mahler volume conjecture Discrete geometry	1939	Proved in dimension 3 (Iriyeh–Shibata 2020)	8	■■■■■
63	Hot spots conjecture Analysis & PDE	1974	Proved for triangles (Judge–Mondal 2022)	7	■■■■■
64	Hadamard conjecture Combinatorics	1893	Smallest unknown order: 668	8	■■■■■
65	Exact value of $R(5, 5)$ Combinatorics	1955	$43 \leq R(5, 5) \leq 46$ (Angelteit–McKay 2024)	8	■■■■■
66	Andrews–Curtis conjecture Topology	1965	Widely believed false; candidate counterexamples remain	8	■■■■■
67	Whitehead asphericity conjecture Topology	1941	Open for over 80 years	8	■■■■■
68	Kaplansky zero-divisor conjecture Algebra & groups	1956	Unit conjecture disproved (Gardam 2021)	8	■■■■■
69	Polynomial Freiman–Ruzsa over $\mathbb{Z}$ Additive combinatorics	1999	Proved over $\mathbb{F}_2^n$ (Gowers–Green–Manners–Tao 2023)	8	■■■■■
70	Sum–product conjecture Additive combinatorics	1983	Exponent $4/3 + c$ (Rudnev–Stevens 2020)	8	■■■■■
71	Log-rank conjecture Complexity	1988	$O(\sqrt{\text{rank}})$ (Lovett; Sudakov–Tomon)	8	■■■■■
72	Polynomial Hirsch conjecture Discrete geometry	1957	Hirsch disproved (Santos 2010); quasi-polynomial bound	8	■■■■■
73	Graph isomorphism in P Complexity	1972	Quasipolynomial time (Babai 2015)	8	■■■■■
74	Furstenberg $\times 2 \times 3$ conjecture Dynamics	1967	Positive-entropy case (Rudolph 1990)	9	■■■■■
75	Zauner’s conjecture (SIC-POVMs) Math. physics	1999	Exact solutions in many dimensions; tied to Hilbert’s 12th	8	■■■■■
76	3D percolation at p_c Probability	1957	Known in 2D and for $d \geq 11$	9	■■■■■
77	Burnside: is $B(2, 5)$ finite? Algebra & groups	1902	Exponents 2,3,4,6 finite; large odd exponents infinite	8	■■■■■
78	Vaught’s conjecture Logic & computability	1961	Counterexamples ruled out in many classes	8	■■■■■
Tier B — strong research problem (excellent paper, notable)					
79	Rota’s basis conjecture Combinatorics	1989	$n - o(n)$ disjoint transversal bases (Pokrovskiy 2020)	7	■■■■■
80	Ryser’s conjecture (Latin transversals) Combinatorics	1967	Partial transversal of size $n - 1$, large n (Montgomery 2023)	7	■■■■■
81	Seymour’s second-neighbourhood conj. Graph theory	1990	Known for tournaments (Fisher 1996)	6	■■■■■
82	Caccetta–Häggkvist conjecture Graph theory	1978	Even the triangle case is open	7	■■■■■
83	Erdős–Gyárfás conjecture Graph theory	1995	Known for claw-free graphs and some planar classes	6	■■■■■
84	Lovász path conjecture Graph theory	1969	Only 5 non-Hamiltonian vertex-transitive graphs known	7	■■■■■
85	Erdős–Turán additive bases Additive combinatorics	1941	Erdős offered \$500	8	■■■■■
86	Happy ending problem (exact) Discrete geometry	1935	$2^{n+o(n)}$ (Suk 2017); exact $2^{n-2} + 1$ open	7	■■■■■
87	Conway’s 99-graph problem Graph theory	1975	Conway offered \$1000	6	■■■■■
88	Moore graph of degree 57 Graph theory	1960	Would have 3250 vertices; cannot be vertex-transitive	7	■■■■■
89	Projective plane of order 12 Combinatorics	1938	Order 10 ruled out by computer (Lam 1989)	8	■■■■■
90	Singmaster’s conjecture Number theory	1971	No number known to appear > 8 times in Pascal’s triangle	6	■■■■■

#	Problem	Since	Best known / status (2026)	D	Fit
91	Brocard's problem $n! + 1 = m^2$ Number theory	1876	Finitely many solutions if abc holds (Overholt)	8	■ ■ ■ ■ ■
92	Perfect cuboid Number theory	1719	Euler bricks exist; no perfect one known	7	■ ■ ■ ■ ■
93	Lehmer's totient problem Number theory	1932	A counterexample needs many prime factors	7	■ ■ ■ ■ ■
94	1/3–2/3 conjecture (posets) Combinatorics	1968	$\delta \geq 0.276$ (Brightwell–Felsner–Trotter 1995)	7	■ ■ ■ ■ ■
95	Gyárfás–Sumner conjecture Graph theory	1975	Known for paths, stars and some trees	8	■ ■ ■ ■ ■
96	Odd covering systems Number theory	1952	Minimum-modulus problem solved (Hough 2015)	7	■ ■ ■ ■ ■
97	Sidon sets: $\sqrt{n} + O(1)$? Additive combinatorics	1941	$\sqrt{n} + 0.998 n^{1/4}$ (Balogh–Füredi–Roy 2023)	7	■ ■ ■ ■ ■
98	Toeplitz inscribed square Discrete geometry	1911	Smooth curves done; rectangles (Greene–Lobb 2020)	7	■ ■ ■ ■ ■
99	Sendov's conjecture (all degrees) Analysis & PDE	1958	True for large degree (Tao 2020)	6	■ ■ ■ ■ ■
100	Busy Beaver $BB(6)$ Logic & computability	1962	$BB(5) = 47,176,870$ (2024, in Coq); $BB(6)$ hides Collatz-like machines	8	■ ■ ■ ■ ■

Statuses reflect my knowledge as of mid-2026 and are brief. Before attacking any of these we should check the current literature; several have had recent partial progress that a one-line summary cannot capture.

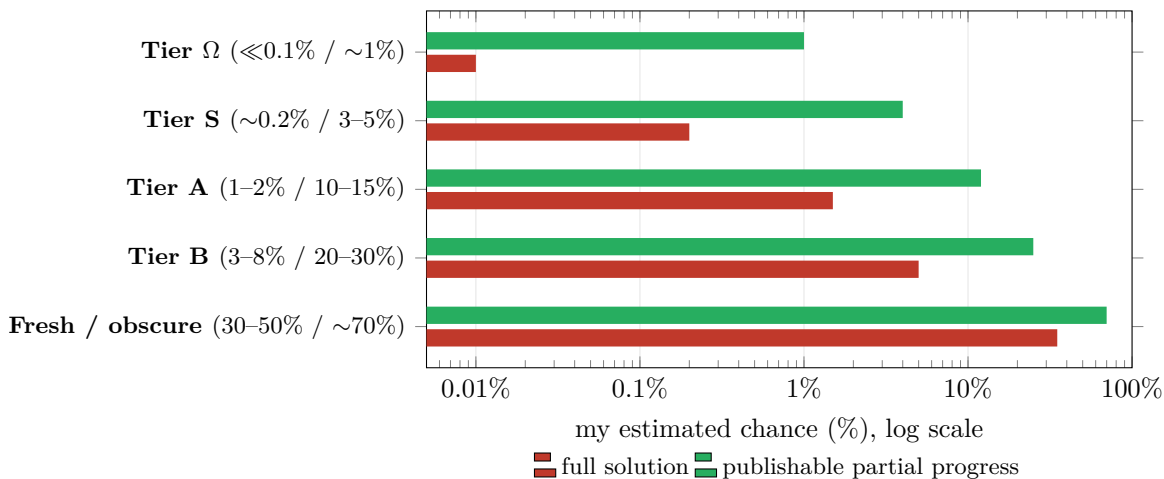
4 My honest opinion: can raw reasoning do it?

You asked me to assume I am smarter than I think. I will take that seriously, but being smart does not change the structure of the problems. Here is what holds regardless of how clever the solver is.

Barriers: proofs that whole approaches cannot work

Problem	Known barrier	So a proof must...
P vs NP	Relativization (1975), natural proofs (1994), algebrization (2008)	use properties of computation that survive oracles and are not “constructive and large”
Twin primes, Goldbach	Selberg’s parity problem: sieves cannot tell 1 prime factor from 2	bring in non-sieve structure, as Friedlander–Iwaniec did for $x^2 + y^4$
Navier–Stokes	Supercriticality; an averaged equation with the same energy identity blows up (Tao)	use the exact algebraic form of the nonlinearity
Riemann Hypothesis	No Hilbert–Pólya operator; the function-field proof (Weil, Deligne) uses geometry with no known analogue over $\mathbb{Q}$	invent the missing geometry (“ $\mathbb{F}_1$ ”) or a genuinely new route
Collatz	Conway: generalised Collatz maps are undecidable	use arithmetic special to $3n + 1$
Frege lower bounds	Feasible interpolation fails for Frege under crypto assumptions	avoid interpolation; perhaps the copying-versus-naming view from our project
Circuit lower bounds	Natural proofs (again)	be non-natural, as Williams’ ACC result was

Odds for one focused, multi-session attack (you + me + Lean, little computation)



What being very capable *does* buy us

- **Breadth:** a systematic search for transplants (page 4) across every field at once.
- **Stamina:** hundreds of small cases, lemmas and reformulations without fatigue.
- **Verification:** Lean catches the plausible but wrong proof, which is the classic failure mode on famous problems.
- **Track record:** last sessions we resolved OEIS A397683 (apparently first) and proved the binary case of our own Lucas conjecture by hand.

What it does *not* remove

- **Barriers.** They hold for every solver, however intelligent.
- **Missing theories.** Langlands-type problems need decades of new concepts, not one argument.
- **Novelty risk.** It is easy to re-prove something known, as happened twice before; a literature check comes first.
- **Overconfidence.** Long “proofs” of famous conjectures are almost always wrong. Lean is our safeguard, and the reason I will not claim a proof I cannot check.

Bottom line. Attacking a Tier Ω problem head-on would be a lottery ticket. A Tier A or B problem in the sweet spot is a real bet with meaningful odds, and even partial progress there (a better constant, a new special case, or a clean reformulation) is publishable. If you want a moonshot, choose one that shares structure with work we have already done: Frege lower bounds connect directly to our naming calculus.

5 My picks: where I would attack

Ordered from the best bet to the boldest. “Lean ready” means a formal statement already exists in the local **formal-conjectures** library, so we can start proving at once.

1. Union-closed sets (Frankl, 1979)

Claim: in every union-closed family, some element lies in at least half of the sets.

Why it fits: Gilmer’s 2022 entropy proof of a constant fraction is about 8 pages, and the problem is pure finite combinatorics. *Lean ready.*

The wall: single-coupling entropy arguments provably stall near 0.382.

First move: two-step couplings, or entropy combined with a compression/shifting argument; test only on tiny families.

2. Černý conjecture (1964)

Claim: a synchronizing n -state automaton has a reset word of length $\leq (n-1)^2$.

Why it fits: a clean statement about words and subsets; the progress has been combinatorial. *Lean ready.*

The wall: 60 years stuck at about $n^3/6$.

First move: a potential function on subsets that provably shrinks every n steps, in the same spirit as our speed-limit potentials.

3. Seymour’s second neighbourhood (1990)

Claim: every oriented graph has a vertex whose second out-neighbourhood is at least as large as its first.

Why it fits: the tournament case was solved by an elegant “median order” idea.

First move: generalise median orders to oriented graphs with few missing edges, then induct on the missing edges.

4. The 1/3–2/3 conjecture (1968)

Claim: every finite non-chain poset has a pair x, y with $\frac{1}{3} \leq \Pr[x < y] \leq \frac{2}{3}$.

Why it fits: a geometric inequality on order polytopes; the 0.276 bound came from Brunn–Minkowski. *Lean ready.*

First move: transplant a sharper log-concavity tool (Lorentzian polynomials, 2020) to beat 0.276.

5. Erdős–Gyárfás (1995)

Claim: every graph of minimum degree 3 contains a cycle whose length is a power of 2.

First move: the structure of a minimal counterexample, plus parity arguments in the cycle space over $\mathbb{F}_2$.

6. Sunflower conjecture (1960)

Claim: k -uniform families with no 3-sunflower have size at most C^k .

Why it fits: 2019 brought a leap through “spread” sets in a short paper; the 3-petal case touches cap sets. *Lean ready.*

First move: combine slice-rank (cap sets) with spread sets.

7. Moonshot: Frege lower bounds

Why us: our naming calculus is a model of the copying-versus-naming gap that *is* Frege vs Extended Frege.

Realistic sub-target: lower bounds for $\text{AC}^0[p]$ -Frege (bounded depth with mod- p gates), open since the 1990s and Tier A in its own right.

8. Warm-up that is ours: the Lucas conjecture

Finish what we started: formalise the binary case $u < 7\varphi^{C/3}$ in Lean, then attack windows of size ≤ 3 by block amortisation. Nobody else is working on it, because we posed it.

Proposed plan for the hard-problem session

1. **You choose** one target from cards 1–6 (my vote: **union-closed sets** or **Černý**).
2. **Literature sweep first** (about 20 minutes of searching), so that we never re-prove a known result.
3. **Transplant brainstorm:** list 10–15 tools from distant fields with the right shape and kill most of them quickly with small, cheap examples.
4. **Aim for a ladder of results:** a new special case, then a better constant, then the full conjecture. Every rung is formalised in Lean before we claim it.
5. **Write it up honestly**, including the dead ends, which other researchers find valuable too.